Revision – Sample Questions

The topics for revision are as follows:

- Linear Programming
- Inventory Modelling
- Simulation
- Transportation Problem
- Forecasting Additive Multiplicative
- Minimum Spanning Tree

I have included sample questions for the first 5 of these topics. The last topic you can find a few practice problems in the tutorial to revise how to compute a minimum spanning tree.

In the exam each question will be worth 25 marks. The questions in this document are not necessarily full questions but are there as examples for you to look at.

- You can also repeat the tutorial questions that are similar.
- Please see the mock paper for a complete idea of the exam.

1. N&F produce two products X and Y. These products are processed through 3 stages A B and C.

   The time of each item at each stage and the available time at each stage is given in table 1.

Times in Hours	X	Y	Available Hours
A	1	3	60
B	4	1	75
C	1	1	27

Table 1

Cost per hour	X	Y
A	3	2
B	3	6
C	4	4
Sales Revenue per Item	24	23

Table 2

a. Use this data and the cost data given in table 2 to determine the optimal production plan for the company if they aim to maximise contribution to profit.

b. Extend the model from part a so that at least 60% of the products are of type X and solve.

c. Extend the model from part b so that the total production cost is less than 400 and solve.

2. MH&C operate a small factory in Deeham. Good receipt is monitored (quality control) by a single worker. However, it is expected that demand is going to increase.

   By constructing and carrying out suitable simulations evaluate the effect of retaining a single worker or employing an additional (part-time) worker.

	Current Demand			
Demand per Hour	5	8	10	12
Frequency (%)	20	40	30	10

	Work Rate of existing worker			
Jobs process per hour	8	9	10	12
Frequency (%)	40	30	20	10

	Expected demand			
Demand per Hour	6	9	12	15
Frequency (%)	50	35	10	5

	Work Rate of new (part time) worker			
Jobs process per hour	4	5	6	7
Frequency (%)	20	60	10	10

For your analysis you can assume that the company makes £20 for every job processed and it costs £20 an hour for the current worker and £10 an hour for a part-time worker.

Simulate single worker current demand

Simulate single worker increased

vs

Simulate single + part-time increased

3. Obtain the Optimal (minimum cost) distribution plan (Source to Customer) for the data

Warehouse	A	B	C
Available Stock	100	200	100

Table 8: Stock availabilities

Supermarket	X	Y	Z
Demand	70	80	110

Table 9: Customer Requirements

The shipping costs per item are given in the table 10 below.

		Supermarkets		
		X	Y	Z
Warehouses	A	5	8	6
	B	3	14	5
	C	2	15	3

Table 10: Transportation Costs warehouse to shop

a. Draw the network representation of this problem
b. Obtain the optimal minimum cost solution
c. Determine the effect on this solution (quantities and cost) if the following (additional) conditions are imposed:

   No more than 60 items may be sent along route BX
   Route BZ cannot be used

d. In addition to c), a distribution centre is added which can act as an intermediate storage location becomes an option for the distribution plan. The costs are given in the table below. Determine if it is worth using and if so, what is the new distribution plan and cost. Note you can still ship directly to the shop from the warehouse.

	Warehouses		
	A	B	C
Distribution Centre	4	2	3

	Supermarkets		
	X	Y	Z
Distribution Centre	3	4	3

4.

a. Determine the Order Quantity that will minimise total yearly costs:

K	£10
D	10000
h	20p
p	80p

b. How would this policy change if quantity discounts were available

$Q \geq 1200$ $p = 79p$

$Q \geq 4000$ $p = 78p$

5. By calculating suitable moving averages determine the trend in the Sales Data. Use this model to estimate the Seasonal Variations (assume an additive model) and hence produce a forecast for the sales over the next year.

Sp	109
Su	115
Au	112
W	125
Sp	149
Su	156
Au	150
W	166
Sp	186
Su	193
Au	196